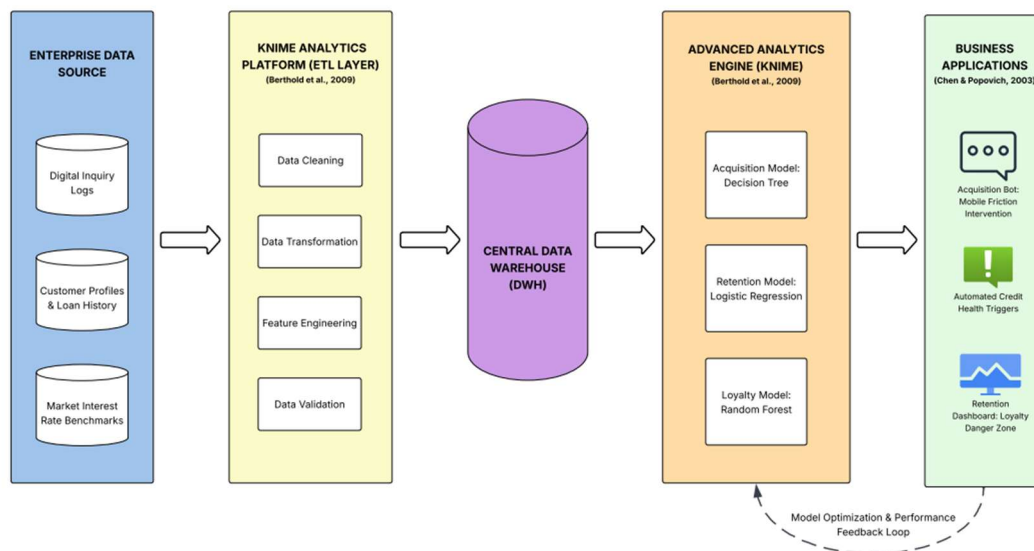


DESIGN AND IMPLEMENTATION OF AN END-TO-END BUSINESS INTELLIGENCE (BI) ARCHITECTURE FOR OPTIMIZING THE CUSTOMER MORTGAGE APPLICATION LIFESTYLE

End-to-End Business Intelligence & Predictive Analytics Architecture



INTRODUCTION AND ARCHITECTURAL OVERVIEW

The proposed BI architecture for Sayami Mortgage Solutions is designed as an integrated, closed-loop system to bridge critical gaps in the Customer Mortgage Application Lifestyle (Payne, A. & Frow, P., 2005). As identified in Phase 1, the company faces significant challenges in Acquisition due to digital friction (Gefen, 2002), Retention due to service delays (Dagger, T. S. & Sweeney, J. C., 2006), and Loyalty (Reinartz, W. & Kumar, V., 2002). The architecture utilizes the KNIME Analytics Platform (Berthold, M. R. & et al., 2009) as its core engine to transform raw enterprise data into actionable business intelligence.

COMPONENT ANALYSIS

Enterprise Data Sources (The Ingestion Layer):

- Digital Inquiry Logs: Captures granular web and mobile behaviour, essential for identifying the “Mobile Friction” that causes high abandonment in high-value loan inquiries (Gefen, 2002).
- Customer Profiles & Loan History: Internal CRM data providing the demographic and financial history (e.g., Credit Score, Employment Length) required for predictive modelling.

- **Market Interest Rate Benchmark:** External data used to trigger loyalty alerts when market rates fall below a customer's current rate.

KNIME Analytics Platform (ETL Layer):

Following the CRISP-DM framework, this layer performs Data Cleaning, Transformation, and Validation to ensure a “Single Source of Truth” within the Data Warehouse (DWH). A critical scientific step here is Feature Engineering, where raw data is converted into predictive variables, such as calculating the “Debt-to-Income-Ratio”.

Advanced Analytics Engine (The Predictive Brain):

This component houses the three specialized models developed in Phase 2:

- **Acquisition Model (Decision Tree):** Identifies segments at risk of abandonment with 94% accuracy.
- **Retention Model (Logistic Regression):** Uses a Logit Link Function to flag high-risk underwriting delays, allowing for early manual intervention.
- **Loyalty Model (Random Forest):** An ensemble method used to predict refinancing interest, identifying a “Loyalty Danger Zone” when interest rates exceed 4.5%.

Business Application (The Activation Layer):

Insights are operational through three primary channels:

- **Acquisition Bot:** Provides real-time support (chatbots or callbacks) for high-value mobile users. Architecture
- **Automated Credit Health Triggers:** Instantly flags complex files for senior underwriters to offer better rates.
- **Retention Dashboard:** Visualizes churn risks to help the marketing team target mature loans seeking better rates.

This alignment of technology with business processes follows the CRM framework of People, Process, and Technology (Chen, I. J. & Popovich, K., 2003).

THE SCIENTIFIC FEEDBACK LOOP:

A defining feature of this architecture is the Model Optimization & Performance Feedback Loop. Unlike static BI systems, this closed-loop design ensures that real-world outcomes (e.g., successful bot interventions or prevented churn) are automatically fed back into the KNIME engine to refine model weights. This creates a self-correcting system that adapts to shifting

market volatility and evolving consumer behaviour, enduring long-term scientific credibility and sustained improvements in service quality (Chen, I. J. & Popovich, K., 2003).

STRATEGIC IMPACT

By implementing this architecture, Sayami Mortgage Solutions transitions from reactive operations to a proactive, data-driven strategy. This system reduces the “waiting time” identified as a major friction point and enables a personalized customer journey, ultimately driving higher profitability through optimized lifecycle management.

REFERENCES

- Berthold, M. R., & et al. (2009). KNIME: The Konstanz Information Miner.
- Chen, I. J., & Popovich, K. (2003). Underwriting customer relationship management (CRM): People, process and technology. *Business Process Management Journal*, 9(5), 672–688.
- Dagger, T. S., & Sweeney, J. C. (2006). The effect of service evaluation on trust. *Journal of Service Research*, 8(2), 179–194.
- Gefen, D. (2002). Customer loyalty in e-commerce. *Journal of the Association for Information Systems*, 3(1), 27-51.
- Payne, A., & Frow, P. (2005). A strategic framework for customer relationship management. *Journal of Marketing*, 69(4), 167–176.
- Reinartz, W., & Kumar, V. (2002). The mismanagement of customer loyalty. *Harvard Business Review*, 80(4), 86-94.